

Kalyani Patrikar

🌐 kalyanipatrikar.github.io

📍 Liverpool, UK | 📞 +447818912 702 | ✉ k.patrikar@liverpool.ac.uk

🔗 kalyanipatrikar | 🌐 [/in/kalyani-patrikar](https://in/kalyani-patrikar) | 🆔 0000-0003-2709-3377

*Materials Scientist and Electrical Engineer, with extensive experience in designing complementary **experimental** and **computational** workflows to probe material properties and device mechanisms.*

EXPERIENCE

Postdoctoral Research Associate, *University of Liverpool*

PRESENT

- Developed physics-based model integrating molecular-dynamics data with charge-transport theory for **charge transport** in mixed conducting polymers
- Developing model for charge transfer at device interfaces from first principles

Early Career Fellow, *Indian Institute of Technology Gandhinagar*

Aug 2022 – Oct 2024

One of the highest postdoctoral level fellowships offered by any Indian Institute

- Demonstrated mechanisms at organic device interfaces by **measurement** of contact resistance in **transistors**, complemented by physics-based **model** and quantum chemical calculations
- Developed **model** for interfacial charge transfer combining quantum chemical calculations with physics-based model, uncovering mechanisms in functionalised **transistors** and Au nanoparticle catalysts
- Designed general purpose **graph attention network** for accurate prediction of diverse properties from molecular structure, robust across dataset size

Senior Research Assistant, *Center for Excellence in Nanoelectronics*

Jul 2013– Dec 2014

- Designed and fabricated oxide nanoparticles embedded organic transistors for toxic gas sensors

Intern, *Indian Nanoelectronics User Program*

May 2010– Jun 2010

- Simulated and carried out self-assembly of Si quantum dots on SiO₂ and HfO₂ thin films for flash memories

SKILLS

Materials Characterisation : XRD (&XRR), AFM (& KPFM), SEM, XPS, Raman Spectroscopy

Electrical Characterisation : I-V, C-V, **Cryogenic I-V**, Four probe measurement

Programming : Python, Bash (& SLURM), C++

Python Libraries : **PyTorch**, RDKit, ASE, MACE, PySCF, Pandas, NumPy, SciPy

ML Frameworks : GNN (GAT, MPNN), CNN, Random Forest

Scientific Software : CP2K, Gaussian16, Quantum Espresso, xTB, Jmol

KEY PUBLICATIONS (COMPLETE LIST AT kalyanipatrikar.github.io/publications)

Journal (*Equal Contribution Authors, † Corresponding Author)

- Kalyani Patrikar[†] and Anirban Mondal[†], "Deciphering the Microscopic Phenomenon Behind Contact Resistances in Interlayer Functionalized Electrodes and Organic Semiconductors", *Phys. Rev. Mater.*, 8, 054606, 2024 [10.1103/PhysRevMaterials.8.054606] < **Physics based model for functionalised organic transistors inferred from device characteristics** >
- Dinesh Trivedi*, Kalyani Patrikar*, Anirban Mondal, "Dataset Size Invariant Graph Attention Network for Prediction of Diverse Molecular Properties", *Mol. Syst. Des. Eng.*, 9, 2024 [10.1039/D4ME00113C] < **General purpose GAT for electronic and optical properties; robust across dataset sizes and origins** >

3. Kalyani Patrikar[†], Valipe Ramgopal Rao, Dinesh Kabra, Anirban Mondal[†], "Understanding the Microscopic Origin of the Contact Resistance at the Polymer–Electrode Interface", *ACS Appl. Mater. Interfaces*, 15 (42), 49427, 2023 [10.1021/acsami.3c10260] < **Measured contact resistance in organic transistors, physics based model of interfacial charge transfer explaining experimental observations** >
4. Kalyani Patrikar, Nakul Jain, Dwaipayan Chakraborty, Priya Johari, Valipe Ramgopal Rao, Dinesh Kabra, "Influence of Pendant Group on Mobility of Organic Thin Film Transistor in Correlation with Reorganization Energy of Molecules", *Adv. Funct. Mater.*, 29 (8), 1805878, 2019 [10.1002/adfm.201805878] < **Transistor characteristics uncover material features influencing charge transport in DA polymers** >

Conference (*Presenting Author)

1. K. Patrikar*, C. Burke, A. Troisi, "Charge Transport Enhancement with Dynamic Disorder", Oral presentation at *European Conference in Molecular Electronics 2025* < **Physics-based model for charge transport in mixed conducting polymers** > Awarded **Best Contributed Talk**

EDUCATION

Doctor of Philosophy

Indian Institute of Technology Bombay

Electrical Engineering

Jan 2015 – Dec 2021

Master of Technology

Indian Institute of Technology Bombay

Materials Science

Jul 2011 – Jun 2013

Bachelor of Technology

Visvesvaraya National Institute of Technology

Metallurgical Engineering & Materials Science

Jul 2007 – May 2011

PHD THESIS

Bulk and Interfacial Charge Transport in Organic Field Effect Transistors

- Demonstrated novel charge doping **mechanism** of self-assembled monolayers at transistor contacts; developed device model based on ab-initio computations, and thinfilm and **device characterisation**
- Reported **highest mobility** for transistors of common DA polymers PTB7 and PTB7Th via device engineering
- Established new **molecular design** paradigm for order-of-magnitude device performance improvement based on **device characterisation**, ab-initio computations, and polaron transport analysis
- Measured, and developed multiscale model to simulate, **cryogenic** characteristics of organic thin film **transistors**

TEACHING & MENTORSHIP

- **Teaching:** Flexible Electronics (**2 Credit Course**) *IIT Gandhinagar*
- **Demonstrator:** Introduction to Spectroscopy, **Chemical Data Discovery and Design** *University of Liverpool*
- **Teaching Assistant:** **Machine Learning**, Physics of Transistors, VLSI Technology, **Microelectronics Lab**, Communication Skills, Thermodynamics *IIT Bombay*
- **Students Mentored:** BTech and MSc. students for thesis project work *IIT Gandhinagar*

EXTRACURRICULAR ACHIEVEMENTS

- Organised and anchored events as the Institute Executive Member in IIT Bombay Academic Council
- Completed high altitude treks in Himalayas, India